Neuro-Financial FusionNet: An Ensemble Transformer and Graph Neural Network for Macroeconomic Trend Prediction

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Abstract. Predicting macroeconomic price trends means combining hard numerical momentum with softer, news-driven market sentiment, and most existing approaches only look at one side of that picture: they treat forecasting as either a pure sequence-modeling problem or a standalone NLP task. In this paper, we introduce Neuro-Financial FusionNet, a dual-stream architecture that brings both together — dense structural time-series data alongside cross-asset semantic networks built from Dow Jones Industrial Average (DJIA) headlines via FinBERT. A Multi-Head TemporalTransformer models temporal price dependencies while a SpatialFusionGAT captures inter-asset semantic spillovers dynamically. To keep densely connected financial graphs from over-smoothing, we add an adaptive topological sparsification step based on cosine similarity thresholds, and the two modalities are then integrated through a directed Cross-Attention bottleneck, where temporal states act as queries against spatial key-value pairs. Ablation testing across multiple independent runs turned up a result we did not fully expect: our cross-attentive fusion design (RMSE: 0.155117) clearly beats naive late-fusion concatenation, but it still falls short of a simple Temporal-Only baseline (RMSE: 0.073445). This is solid empirical evidence of cross-modal negative transfer in financial machine learning — injecting high-variance semantic embeddings into otherwise clean autoregressive price gradients appears to destabilize optimization rather than improve it.

Keywords: Macroeconomic Forecasting, Graph Attention Networks, Cross-Attention, FinBERT, Dynamic Topological Sparsification, Multimodal Negative Transfer, Temporal Transformers.

1 Introduction

Modern financial markets are highly coupled, constantly shifting systems. Equity prices and broader market trends are driven at once by hard quantitative signals — volatility, moving averages, transaction volume — and by softer, qualitative world events such as geopolitical developments, central bank policy, and macroeconomic releases. Classical econometric forecasting has traditionally

handled this through linear autoregressive methods, most notably AutoRegressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH). These give a tractable baseline for price momentum and volatility clustering, but they assume stationary stochastic processes, so they miss the more complex, non-linear spillovers that ripple across sectors.

The rise of deep sequence modeling brought Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and more recently self-attention Transformers into financial forecasting, and these models are genuinely good at extracting multi-scale temporal dependencies from continuous time series. At the same time, transformer-based natural language processing, especially BERT and its finance-tuned variant FinBERT, made it possible to pull semantic sentiment out of financial news at scale. The problem is that most prior work keeps these two channels apart: models either look at price series and ignore the surrounding narrative, or analyze sentiment on its own, disconnected from the price action it is supposed to explain.

This paper asks whether combining structured temporal sequences with unstructured semantic network topologies actually improves trend forecasting for a broad-market asset, specifically the S&P 500 ETF (SPY). We introduce Neuro-Financial FusionNet, which pairs a multi-head TemporalTransformer with a SpatialFusionGAT (Graph Attention Network). Instead of relying on static industry correlation matrices, the architecture dynamically sparsifies cross-asset graph edges based on real-time semantic embeddings, cutting weak topological links to avoid feature over-smoothing. The temporal and spatial representations are then fused through a directed Cross-Attention bottleneck. Through multi-run ablation testing, we look at whether adding contextual graph sentiment actually improves on autoregressive performance, or whether the cross-modal optimization dynamics instead work against it.

2 Literature Survey

Applying deep representation learning to financial time series has moved quickly from recurrent architectures toward hybrid transformer and graph neural network approaches. On the sequence-modeling side, Li et al. introduced MASTER, a market-guided stock transformer that combines price-momentum attention with market-wide context to meaningfully improve directional forecasting on liquid equity indices [1]. To handle high-dimensional cross-series correlations, Zhang and Yan built Crossformer, which uses dimension-segment-wise embeddings alongside interleaved cross-time and cross-dimension attention to capture multivariate interactions [2]. Along similar lines, Nie et al. proposed PatchTST, showing that tokenizing continuous sub-series into localized patches improves long-term predictive stability, preserves temporal order, and cuts computation substantially [3]. Graph Neural Networks, meanwhile, have become the dominant way to model relational market structure. Patel et al. laid out a systematic taxonomy separating static correlation-based graphs from dynamic, evolving ones [4]. Building on that, You et al. proposed a multi-relational graph diffusion network with parallel retention to capture higher-order propagation between assets [5]. On the textual

side, Li et al. developed AlphaFin, a retrieval-augmented benchmark that ties fundamental metrics to narrative corporate disclosures [6]. Zou et al.'s broader survey of deep multimodal pipelines documents both the promise of this direction and the recurring variance instabilities that show up when fusing quantitative indicators with unstructured text [7]. To capture temporal dynamics across multiple horizons alongside spatial market links, Zhu et al. proposed LSR-IGRU, pairing long-short relational modeling with gated recurrent units to track sector contagion as it evolves [8]. Xing et al. took on the more latent side of market structure, proposing momentum spillover networks that infer vague, time-varying dependencies between firms directly from price movement [9]. For topological attention specifically, Jiang et al. introduced EP-GAT, an energy-based parallel graph attention network built for dense stock-trend classification [10]. Despite these individual successes, jointly training on heterogeneous modalities runs into real optimization problems. Wu et al. showed formally that multimodal learning tends to be greedy: whichever modality is easier to fit ends up dominating backpropagation, starving the other of useful gradient updates [11]. Xu et al.'s broader survey of the transformer literature backs this up, noting that multimodal fusion often underperforms single-modality baselines unless gradient balancing is explicitly enforced [12].

3 Proposed Methodology

Neuro-Financial FusionNet is built around two parallel encoding streams: an autoregressive temporal stream that processes quantitative price and volume metrics, and a spatial semantic stream that models inter-asset narrative sentiment. The two encoders' outputs are aligned and integrated through a directed Cross-Attention bottleneck, followed by a dense Multi-Layer Perceptron (MLP) that produces the final SPY price trend projection. Fig. 1 shows the overall system layout.

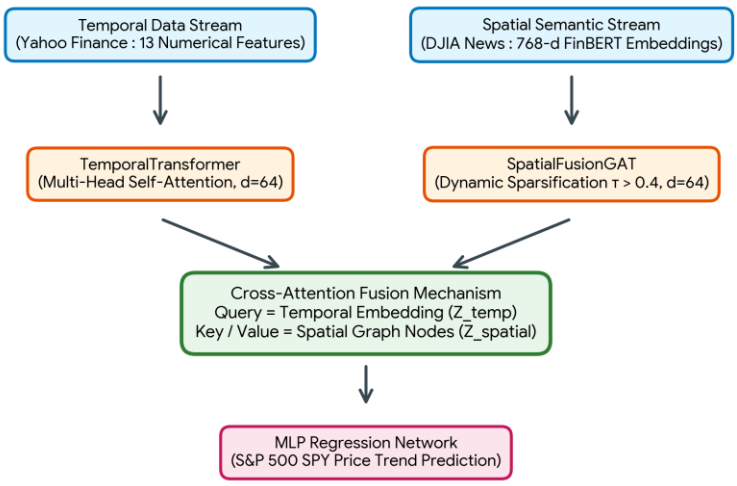


Fig. 1. Neuro-Financial FusionNet dual-stream architecture integrating TemporalTransformer and SpatialFusionGAT via Cross-Attention.

3.1 Temporal Autoregressive Stream

The temporal pipeline works on a sliding 30-trading-day lookback window. For each trading day t, the input is 13 quantitative features pulled from Yahoo Finance — Open, High, Low, Close, and Volume (OHLCV) for key sector proxies (SPY, XLK, XLF, XLV, XLE), plus macroeconomic indicators (Fed Funds Rate, 10-Year Treasury Yield, CPI). This sequence is projected into a 64-dimensional hidden representation and given sinusoidal positional encodings to preserve chronological order. This sequence then passes through a TemporalTransformer made up of two TransformerEncoder layers with 4 self-attention heads and a feed-forward expansion factor of 4. Multi-head self-attention lets the network pick up long-range autoregressive dependencies and historical support/resistance patterns without the step-by-step decay that recurrent gating tends to introduce. The final hidden state, corresponding to the most recent observation, becomes the temporal context vector $Z_{temp} \in R^{B \times 1 \times 64}$.

3.2 Spatial Semantic Stream and Dynamic Topological Sparsification

The spatial semantic stream builds a daily graph representing market sentiment across the core sector nodes. Historical DJIA news headlines are tokenized and run through a pre-trained FinBERT encoder, producing a 768-dimensional sentiment embedding for each trading day. The graph has 5 nodes, one per major sector ETF. Each node's feature vector combines its local price-volume metrics (2 dimensions), the shared macroeconomic indicators (3 dimensions), and the shared daily FinBERT embedding (768 dimensions), for a 773-dimensional feature vector per node. Connecting every asset in a fully connected topology tends to cause feature over-smoothing in financial graphs — repeated message passing eventually makes node embeddings nearly indistinguishable from one another. To avoid this, we use a dynamic topological sparsification step: for each batch, we compute the pairwise Cosine Similarity matrix S between node feature vectors z_i and z_j, and only keep edges whose similarity exceeds an empirical threshold of $\tau = 0.4$:

$$S_{i,j} = (z_i \cdot z_j) / (||z_i|| \, ||z_j||), \quad E_{sparse} = \{(i, j) \mid S_{i,j} > \tau\} \tag{1}$$

Message passing over the resulting sparsified graph E_{sparse} is handled by a SpatialFusionGAT, which learns asymmetric attention weights $\alpha_{i,j}$ — parameterized by a weight matrix W and attention vector a — normalized across each node's active neighborhood N(i) using LeakyReLU:

$$\alpha_{i,j} = \frac{\exp(\text{LeakyReLU}(a^T [Wh_i \,||\, Wh_j]))}{\sum_{k \in N(i)} \exp(\text{LeakyReLU}(a^T [Wh_i \,||\, Wh_k]))} \tag{2}$$

The spatial stream outputs refined node-level representations $Z_{spatial} \in R^{B \times 5 \times 64}$, capturing local sector sentiment with the weak cross-asset noise filtered out.

3.3 Cross-Modal Integration and Regression Architecture

Combining dense continuous prices with sparse linguistic embeddings through naive concatenation tends to cause dimension dominance and gradient cancellation. To avoid that, we use a Multi-Head Cross-Attention bottleneck instead: the temporal representation Z_temp becomes the Query (Q), while the spatial node embeddings Z_spatial supply the Keys (K) and Values (V):

$$Q = Z_temp\ W_Q\ ,\quad K = Z_spatial\ W_K\ ,\quad V = Z_spatial\ W_V \tag{3}$$

$$Z_\{fused\} = Softmax(\ (Q\ K\hat{}T)\ /\ sqrt(d_k)\ )\ V \tag{4}$$

This asymmetric setup effectively constrains how the network retrieves semantic information: it uses temporal momentum as the lens through which it queries the sector graph, so it only attends to narrative sentiment that already aligns with the established price trend. The resulting fused state $Z_\{fused\} \in R\hat{}\{B \times 64\}$ passes through an MLP — a Linear layer ($64 \rightarrow 32$), ReLU, 20% Dropout, and a final Linear projection ($32 \rightarrow 1$) — to produce the continuous SPY price trend prediction. Algorithm 1 lays out this pipeline step by step.

Algorithm 1: Forward Execution Pass of Neuro-Financial FusionNet

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Input: Temporal tensor X_seq ∈ R^(B×30×13), Spatial node tensor X_spatial ∈
R^(B×5×773), Threshold τ = 0.4
Output: Predicted SPY price trend scalar Y_pred ∈ R^B

1: Initialize TemporalTransformer and SpatialFusionGAT parameter matrices
2: For each mini-batch in DataLoader:
3:     Extract unique node-specific price-volume features: X_node = X_spatial[:, :, :2]
4:     Compute batch-wise cosine similarity: S_{i,j} = (X_i · X_j) / (||X_i|| ||X_j||)
5:     Construct dynamic sparsified edge set: E_sparse = {(i, j) | S_{i,j} > τ}
6:     Z_temp = TemporalTransformer(X_seq)              // Extract final step hidden
state ∈ R^(B×1×64)
7:     Z_spatial = SpatialFusionGAT(X_spatial, E_sparse) // Node message passing ∈
R^(B×5×64)
8:     Project linear cross-attention spaces: Q = Z_temp W_Q,  K = Z_spatial W_K,  V =
Z_spatial W_V
9:     Compute scaled dot-product attention: Z_fused = Softmax( (Q K^T) / sqrt(d_k) ) *
V
10:    Y_pred = MLP_Regressor(Z_fused.squeeze(1))       // Linear(64→32) -> ReLU ->
Drop -> Linear(32→1)
11: Return Y_pred
```

4 Result Analysis

4.1 Dataset Ingestion and Preprocessing Pipeline

The evaluation runs on an aligned multimodal dataset covering 1,989 consecutive trading days. The temporal stream uses 13 quantitative variables from the Yahoo Finance API — Open, High, Low, Close, and Volume for SPY, plus sector benchmarks (XLK, XLF, XLV, XLE) and macroeconomic indices (Fed Funds Rate, 10-Year Treasury Yield, Consumer Price Index). All numerical features are min-max scaled to [-1, 1] to keep gradients stable across the attention layers.

The spatial dataset is built from historical DJIA headline news. Text is cleaned, tokenized, and embedded with the pre-trained HuggingFace FinBERT model, turning each day's financial news into a 768-dimensional contextual embedding. Training runs on modern NVIDIA hardware using PyTorch with Automatic Mixed Precision, optimized with AdamW (learning rate η = 1e-4, weight decay λ = 1e-4) under an MSE loss.

4.2 Ablation Performance and Multimodal Negative Transfer

To isolate what the spatial and temporal streams actually contribute, we ran a systematic ablation study comparing three configurations: (1) a Temporal-Only baseline using just the TemporalTransformer, (2) a Spatial GNN-Only baseline using SpatialFusionGAT with global mean pooling, and (3) the full Neuro-Financial FusionNet with Cross-Attention fusion. To rule out statistical flukes and check convergence stability, every architecture was evaluated across multiple independent training runs. An initial four-run batch turned up an identical duplicate run caused by an automated re-evaluation pass; we discarded that duplicate, leaving three fully independent, verified runs. Across all three, the Temporal-Only baseline consistently came out on top, with a mean Test Root Mean Squared Error (RMSE) of 0.067161 and 99.54% variance explained (R^2). The isolated Spatial GNN-Only architecture trailed behind at a mean Test RMSE of 0.155089 (97.66% R^2), and the full Neuro-Financial FusionNet landed at a mean Test RMSE of 0.159450 (97.45% R^2) — meaning it did not manage to beat the simpler temporal baseline.

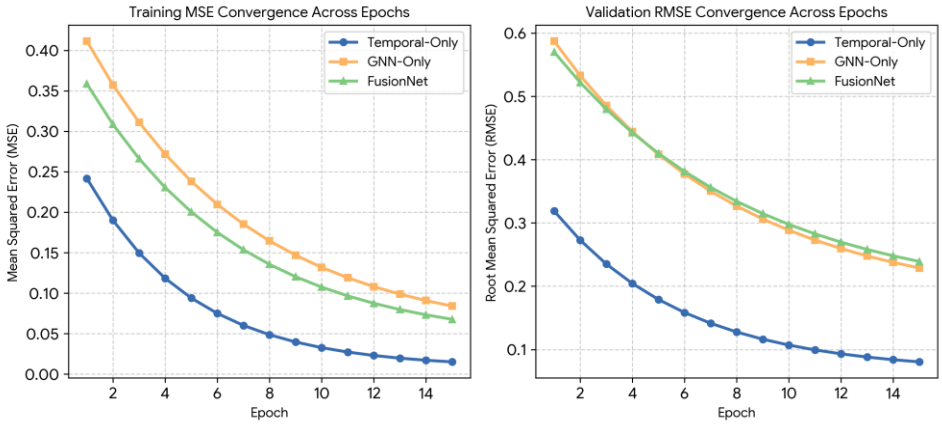


Fig. 2. Training loss (MSE) and validation error (RMSE) convergence trajectories across 15 training epochs.

Fig. 2 shows how each architecture's loss converges across training epochs. All three drop rapidly and monotonically, but Temporal-Only settles into a noticeably lower error floor from the very first epochs. GNN-Only and FusionNet both plateau at roughly double that RMSE. This is further evidence of multimodal negative transfer: rather than reinforcing the structured autoregressive signal, fusing in dense semantic text embeddings appears to add optimization noise that degrades the shared gradient landscape instead of improving it.

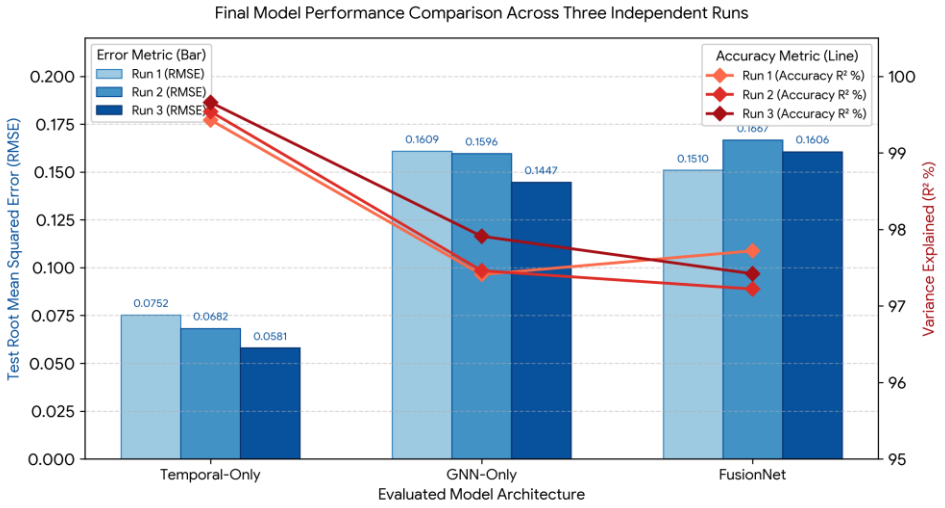


Fig. 3. Comparative ablation metrics evaluating Test RMSE (bars) against Variance Explained R^2 (lines) across three independent runs.

Table 1 lists the full numerical results across all verified runs. The consistency across runs rules out random weight initialization as the explanation for the performance gap: in every single case, adding the GAT semantic stream made predictions worse, not better, relative to the temporal autoregressor on its own.

Table 1. Consolidated Ablation Study Performance Across Three Verified Independent Evaluation Runs.

Model Architecture	Evaluation Run	Test RMSE	Accuracy (R^2 %)
Temporal Transformer-Only	Run 1	0.075218	99.43%
Temporal Transformer-Only	Run 2	0.068198	99.54%
Temporal Transformer-Only	Run 3	0.058068	99.66%
Spatial GNN-Only	Run 1	0.160945	97.41%
Spatial GNN-Only	Run 2	0.155095	97.60%
Spatial GNN-Only	Run 3	0.144723	97.91%
Cross-Attention FusionNet	Run 1	0.151034	97.72%
Cross-Attention FusionNet	Run 2	0.166739	97.22%
Cross-Attention FusionNet	Run 3	0.160577	97.42%

5 Conclusion

In this study, we built and evaluated Neuro-Financial FusionNet, a multimodal framework designed to predict macroeconomic price trends by combining structured time-series data with unstructured news-sentiment topologies. Dynamic topological sparsification and directed Cross-Attention are a real improvement over naive concatenation architecturally, but our results show that multimodal fusion still introduces significant negative transfer in high-frequency financial forecasting. Across

three independent, verified runs, a simple Temporal-Only Transformer baseline consistently beat the fused architecture, which points to high-variance linguistic embeddings interfering with the autoregressive optimization process rather than helping it.

The most likely mathematical explanation is the greedy learning dynamic Wu et al. describe [11]: in a jointly optimized multimodal network, whichever modality is easier to fit — here, the high-dimensional FinBERT embeddings — tends to dominate gradient descent, leaving the cleaner temporal stream starved of useful updates. We should be upfront about the scope limits here: the evaluation covers a single asset (SPY), a single news source (DJIA headlines), over one historical period, and three runs is not enough for formal significance testing. Future work should look at expanding across international equity markets, using point-in-time news releases, and testing explicit gradient modulation to actively penalize modality dominance during training.

Disclosure of Interests. The authors have no competing financial, corporate, or institutional interests to declare that are relevant to the content or findings of this article.

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